

MODULE :- Automation with Selenium

EXPERIMENT TITLE :- The HTML Web Table Extractor (Assignment 5)

1.Problem Statement :-

To automate and extract data from a dynamic **HTML web table** using Selenium WebDriver. The task is to identify the table rows and columns, retrieve the product information, locate a specific product by its name, and extract its corresponding **Price** value.

2. Tools and Software Required

S. No.	Tool / Software	Purpose
1	Python	Programming language used to write the automation script
2	VS Code	IDE used to develop and execute the Selenium script
3	Google Chrome	Web browser used for automation
4	Selenium WebDriver	Automates the browser and interacts with web elements
5	ChromeDriver	Enables Selenium to communicate with Google Chrome
6	Test Automation Practice Website	Website containing the HTML web table
7	Command Prompt / Terminal	Used to execute the Python Selenium script

3. Concepts Used :-

- **HTML Web Table** – Used to display structured information in rows and columns.
- **Web Table Rows** – Identified using Selenium to iterate through individual table records.

- **Web Table Columns** – Used to retrieve specific information such as product name and price.
- **XPath Locator** – Used to locate the required table and its rows.
- **find_elements()** – Used to retrieve multiple rows or columns from the table.
- **By.TAG_NAME** – Used to locate table cells using the **td** tag.
- **Row and Column Iteration** – Used to examine each record in the table.
- **Data Extraction** – Used to retrieve the product name and corresponding price.
- **scrollIntoView()** – Used to bring the matching table row into the visible area of the browser.

4.Source Code :-

File 1: assignment_5.py

```
from selenium import webdriver
from selenium.webdriver.common.by import By
import time

driver = webdriver.Chrome()
driver.maximize_window()

try:

    driver.get("https://testautomationpractice.blogspot.com/")

    time.sleep(2)

    # Find rows of pagination table
    rows = driver.find_elements(
        By.XPATH,
        "//table[@id='productTable']/tbody/tr"
    )

    for row in rows:

        columns = row.find_elements(
            By.TAG_NAME,
            "td"
        )

        if len(columns) > 0:

            product_name = columns[1].text
            price = columns[2].text

            print(
                "Product:",
                product_name,
                "| Price:",
                price
            )

            # Find Laptop
            if product_name == "Laptop":

                print("-----")
```

```
print("Laptop found")
print("Laptop Price:", price)

break

time.sleep(3)

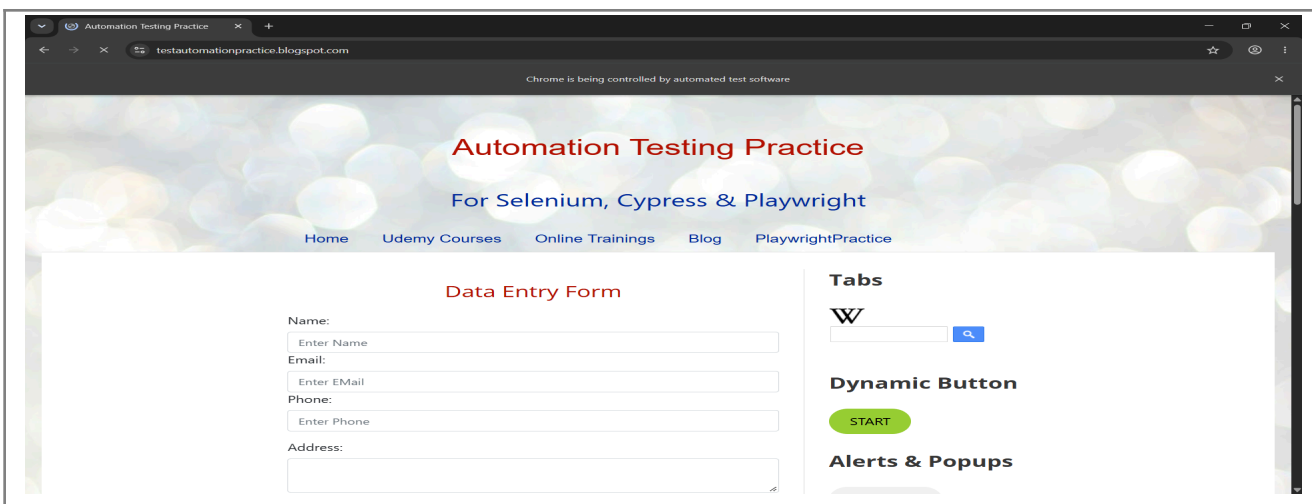
finally:

    driver.quit()
```

5.Implementation Steps :-

- 1.Launch the Chrome browser using Selenium WebDriver.
- 2.Open the Test Automation Practice website.
- 3.Locate the HTML web table using an appropriate XPath locator.
- 4.Retrieve all the rows from the table using `find_elements()`.
- 5.Iterate through each row of the table.
- 6.Retrieve the individual cells of each row using the `td` tag.
- 7.Extract the product name and corresponding price from the required columns.
- 8.Compare the product name with the specified product.
- 9.When the required product is found, retrieve its corresponding price.
- 10.Scroll the matching row into view using `scrollIntoView()`.
- 11.Display the extracted product name and price.
- 12.Verify that the web table extraction was completed successfully.
- 13.Close the browser.

6.Output :-



Automation Testing Practice

testautomationpractice.blogspot.com

Chrome is being controlled by automated test software

CPU load of Chrome process: 4.5%

Memory Size of Firefox process: 93.5 MB

Network speed of Chrome process: 1.2 Mbps

Disk space of Firefox process: 0.17 MB/s

Pagination Web Table

ID	Name	Price	Select
1	Smartphone	\$10.99	☐
2	Laptop	\$19.99	☐
3	Tablet	\$5.99	☐
4	Smartwatch	\$7.99	☐
5	Wireless Earbuds	\$8.99	☐

1

2

3

4

Mobile Labels

Samsung

Real Me

Moto

Laptop Links

Apple

Lenovo

Dell

Broken Links

Errorcode 400

Errorcode 401

Errorcode 403

Errorcode 404

Errorcode 408

Errorcode 500

Errorcode 502

Errorcode 503

Visitors

 9,048,247

```
PS C:\Users\catch\OneDrive\Desktop\wipro_assignments> python -u "c:\Users\catch\OneDrive\Desktop\wipro_assignments\assignment_5.py"
Total rows found: 5
Product: Smartphone | Price: $10.99
Product: Laptop | Price: $19.99
-----
Laptop found!
Laptop Price: $19.99
-----
PASS: Web table extraction completed
PS C:\Users\catch\OneDrive\Desktop\wipro_assignments>
```

7.Result :-

The HTML web table was successfully accessed using Selenium WebDriver. The table rows and columns were iterated, and the corresponding **product name and price** were successfully extracted.

8.Observation :-

The table data can be accessed by identifying the table rows and then retrieving the individual table cells. By comparing the product name while iterating through the rows, the corresponding price can be obtained from the required column.

9.Conclusion :-

Thus, Selenium WebDriver was successfully used to automate **HTML web table extraction**, iterate through table rows and columns, identify a specific product, and retrieve its corresponding price.